

Problem 5

Problem 1 Select the meaning of $\lim_{x \rightarrow \infty} f(x) = 6$. Support your explanation graphically.

- (a) As x becomes arbitrarily negatively large, $f(x)$ approaches 6.
- (b) As x becomes arbitrarily positively large, $f(x)$ approaches 6.
- (c) As x approaches 6, $f(x)$ becomes arbitrarily negatively large.
- (d) As x approaches 6, $f(x)$ becomes arbitrarily positively large.

Explanation. As x becomes arbitrarily positively large, $f(x)$ approaches 6. Here are two examples of what this function might look like:

